

REMARKS ON SIX-TANGENTIAL CONICS TO SMOOTH PLANE SEXTICS

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ABSTRACT.

Let k be a positive integer. We consider the complete graph with vertex set $V_k = \{1, \dots, k\}$ and edge set

$$E_k := \{[i, j] \mid i, j \in V_k, i < j\}.$$

Let m be a positive integer, and we put

$$M := \{-m, -m+1, \dots, m\}.$$

Let $\mathcal{W}_k$ be the set of weights $w: E_k \rightarrow M$, that is, $\mathcal{W}_k$ is the set of weighted complete graphs on V_k with weights in M . We let the symmetric group $\mathfrak{S}_k$ act on V_k from the right. Then $\mathcal{S}_k$ acts on $\mathcal{W}_k$ from the left. We define Sw_k to be the multiplicative group of all mappings $s: V_k \rightarrow \{\pm 1\}$, which acts on $\mathcal{W}_k$ by switching, that is,

$$(sw)([i, j]) = s(i)s(j)w([i, j])$$

for $s \in \text{Sw}_k$ and $[i, j] \in E_k$. Then the semidirect product

$$G_k := \text{Sw}_k \rtimes \mathfrak{S}_k$$

acts on $\mathcal{W}_k$ from the left. The purpose of this section is to describe an efficient algorithm for obtaining the list of orbits of the action of G_k on $\mathcal{W}_k$.

First, we introduce a total order on the set E_k of edges by

$$[i, j] < [i', j'] \iff j < j' \text{ or } (j = j' \text{ and } i < i').$$

Then we express elements w of $\mathcal{W}_k$ by arranging the values $w([i, j])$ in this order:

$$w = (w([1, 2]), w([1, 3]), w([2, 3]), w([1, 4]), \dots).$$

Let w^{ab} denote the vector obtained from w by taking the absolute value of each component. We put $M^{\text{ab}} := \{0, 1, \dots, m\}$. Then we have a mapping

$$(0.1) \quad \mathcal{W}_k \longrightarrow \mathcal{W}_k^{\text{ab}} := (M^{\text{ab}})^{E_k}, \quad w \longmapsto w^{\text{ab}}.$$

Notice that $\mathfrak{S}_k$ acts on $\mathcal{W}_k^{\text{ab}}$, and the mapping (0.1) is $\mathfrak{S}_k$ -equivariant. Moreover, the action of Sw_k on $\mathcal{W}_k$ preserves each fiber of (0.1). We then define a total order on $\mathcal{W}_k$ by

$$w < w' \iff w^{\text{ab}} <_{\text{lex}} w'^{\text{ab}} \text{ or } (w^{\text{ab}} = w'^{\text{ab}} \text{ and } w <_{\text{lex}} w').$$

We say that $w \in \mathcal{W}_k$ is G_k -*minimal* if w is minimal in the orbit $G_k w$. We obtain the list of G_k -orbits in $\mathcal{W}_k$ by enumerating all G_k -minimal elements in $\mathcal{W}_k$. By definition, we have the following:

$$\begin{aligned} w^{\text{ab}} \text{ is minimal with respect to } <_{\text{lex}} \text{ in the } \mathfrak{S}_k\text{-orbit} \\ w \text{ is } G_k\text{-minimal} \iff \mathfrak{S}_k w^{\text{ab}} \subset \mathcal{W}_k^{\text{ab}}, \text{ and } w \text{ is minimal with respect to} \\ <_{\text{lex}} \text{ in the } \text{Sw}_k\text{-orbit } \text{Sw}_k w. \end{aligned}$$

By this observation, to determine whether an element $w \in \mathcal{W}_k$ is G_k -minimal, it suffices to make at most $k! + 2^k$ comparisons, which is much smaller than the size $2^k k!$ of G_k .

Suppose that $k' < k$. By taking the prefix of length $k'(k'-1)/2$ of each $w \in \mathcal{W}_k$, we obtain a mapping

$$\mathcal{W}_k \longrightarrow \mathcal{W}_{k'}, \quad w \longmapsto w|_{k'}.$$

It is obvious that, if $w'|_{k'} < w|_{k'}$ in $\mathcal{W}_{k'}$, then we have $w' < w$ in $\mathcal{W}_k$. On the other hand, we have a natural embedding $G_{k'} \hookrightarrow G_k$ denoted by $g \mapsto \tilde{g}$ by

$$i^{\tilde{\tau}} := \begin{cases} i^{\tau} & \text{if } i \leq k', \\ i & \text{if } i > k', \end{cases} \quad \tilde{s}(i) := \begin{cases} s(i) & \text{if } i \leq k', \\ i & \text{if } i > k', \end{cases}$$

for $\tau \in \mathfrak{S}_{k'}$ and $s \in \text{Sw}_{k'}$. Then we have

$$(\tilde{g}w)|_{k'} = g(w|_{k'})$$

for $\tilde{g} \in G_{k'}$ and $w \in \mathcal{W}_k$. Therefore we obtain:

Lemma 0.1. *Suppose that $w \in \mathcal{W}_k$ is G_k -minimal. Then $w|_{k'}$ is $G_{k'}$ -minimal. $\square$*

Using this lemma, we can construct the list of G_k -minimal elements in $\mathcal{W}_k$ by induction on k .

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